

MINISTRY OF EDUCATION, CULTURE AND RESEARCH OF THE REPUBLIC OF MOLDOVA

Technical University of Moldova
Faculty of Computers, Informatics and Microelectronics
Department of Software and Automation Engineering

Iulian Pavlov, FAF-241

Report

Laboratory work nr. 1

The OS as a Resource Manager

Checked by:
Reițman Patricia, *University Assistant*
Department of Software and Automation Engineering,
FCIM Faculty, UTM

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Part 1 – Files and Directories (File Management)

```
$ ls -l note.txt
```

```
Permissions Size User      Date Modified Name
.rw-r--r--    24 sunflower 28 Sep 16:39  note.txt
```

```
$ chmod 600 note.txt
$ ls -l note.txt
```

```
Permissions Size User      Date Modified Name
.rw-------    24 sunflower 28 Sep 16:39  note.txt
```

Observations:

1. **Who owns the files you just created, and which group?**
   The files are owned by the user `sunflower` and belong to the primary group `sunflower`.

2. **What do the ten characters at the start of `ls -l` mean for `note.txt` after `chmod 600`?**
   The permission string `.rw-------` (representing standard UNIX `-rw-------`) breaks down into:

   - Position 1 (`-`): denotes a regular file.
   - Positions 2–4 (`rw-`): owner permissions; user `sunflower` has read (`r`) and write (`w`) permissions without execute (`-`).
   - Positions 5–7 (`---`): group permissions; group members have no access rights.
   - Positions 8–10 (`---`): others permissions; all other system users have no access rights.

3. **Name two directories directly under `/` and say, in one line each, what they hold.**

   - `/etc`: stores host-specific, system-wide configuration files and startup scripts.
   - `/home`: stores personal home directories, personal data, and user configurations.

Part 2 – Processes (Process Management)

```
$ cat /proc/$PID/status | head -20
```

```
Name:   sleep
Umask:  0022
State:  S (sleeping)
Tgid:   87978
Ngid:   0
Pid:    87978
PPid:   7375
TracerPid:      0
```

```
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 998 1000
NStgid: 87978
NSpid:  87978
NSpgid: 87978
NSsid:  7375
Kthread:        0
VmPeak:     6016 kB
VmSize:     6016 kB
VmLck:         0 kB
```

Observations:

1. **What is the PID of process number 1, and what is it (look it up)?**
   The PID of process number 1 is `1`, executing `/sbin/init` (systemd). It is the first user-space process spawned directly by the Linux kernel during system boot, serving as the ancestor of all subsequent processes and system services.

2. **Roughly how many processes were running on your idle VM?**
   According to `top` and `ps aux | wc -l`, approximately 357 to 359 processes were running concurrently on the idle system, with roughly 350 of them in a sleeping state.

3. **From `/proc/<pid>/status`, what does the "State:" line say for your sleeping process?**
   For the background `sleep 300 &` process, the line reports:

   ```
   State:  S (sleeping)
   ```

   This indicates interruptible sleep: the process is dormant, waiting for a timer event to expire without consuming CPU cycles.

Part 3 – Memory (Memory Management)

```
$ free -h
```

```
               total        used        free      shared buff/cache   available
Mem:            15Gi       3.7Gi        9Gi       552Mi       2.9Gi        11Gi
Swap:           30Gi          0B        30Gi
```

```
$ grep VmRSS /proc/$PID/status
```

```
VmRSS:      2444 kB
```

Observations:

1. **How much total RAM does the VM have, and how much is free?**
   The system has approximately 15 GiB of total RAM (`15,993,828 kB` in `/proc/meminfo`). At the time of execution, roughly 9.97 GiB (`10,457,120 kB`) was unallocated, and approximately 11 GiB was available when counting reclaimable buffers and cache.

2. **What is swap, and how much is configured?**
   Swap is secondary storage space (such as a disk partition or a zram device) used as an overflow buffer for physical RAM when memory demand peaks. The system has 30 GiB of total swap configured (including a 15.3 GiB `zram0` compressed paging device), with 0 B in use.

3. **How much resident memory (VmRSS) does a bare `sleep` process use, and does that surprise you?**
   The process uses `2444 kB` (approximately 2.4 MB) of resident set size. It is negligible overall, but surprisingly non-zero for a process that computes nothing; the overhead exists because the OS must load the dynamic linker (`ld.so`), the standard C library (`libc.so`) and kernel memory structures before the process pauses.

Part 4 – Devices and Storage (I/O Management)

```
$ lsblk
```

```
NAME        MAJ:MIN RM   SIZE RO TYPE  MOUNTPOINTS
zram0       252:0    0  15.3G  0 disk  [SWAP]
nvme0n1     259:0    0 476.9G  0 disk
|-nvme0n1p1 259:1    0     2G  0 part  /boot
`-nvme0n1p2 259:2    0 474.9G  0 part
  `-root    253:0    0 474.9G  0 crypt /home
                                       /var/cache/pacman/pkg
                                       /var/log
                                       /
```

Observations:

1. **Which device is your root filesystem "/" mounted on?**
   The root filesystem `/` is mounted on the encrypted device `/dev/mapper/root`, which resides on the physical NVMe SSD partition `/dev/nvme0n1p2`.

2. **Give one entry from `/dev` and say what real thing it stands for.**
   The entry `/dev/dri` represents the Direct Rendering Infrastructure, which interfaces with the physical graphics hardware (GPU) to enable hardware-accelerated video output and 3D rendering.

3. **In one sentence: what does "everything is a file" mean, based on what you saw?**
   "Everything is a file" means the operating system exposes hardware devices, memory, storage disks and system processes through one consistent tree of file descriptors, so user-space programs can use the same `open`, `read` and `write` system calls for all of them.

Conclusion

In this laboratory work, I explored the operating system as a resource manager that arbitrates files, processes, memory and devices. I inspected file ownership, permissions and the directory hierarchy using `ls` and `chmod`, and monitored process states through `ps`, `top` and `/proc`. I then checked physical memory, swap and per-process resident memory with `free` and `VmRSS`, and finally examined disks, partitions and mounted filesystems with `lsblk` and `/dev`. Together these exercises showed how the kernel presents each resource through a uniform, file-based interface.